

High-Protein Food Value Analysis: A Data-Driven Study of Protein Efficiency Across Branded Foods

Goal:

My goal for this project was to gain a deeper understanding on branded foods that hold a High protein profile and how trends answer questions. The foods lay across different categories which provide an interesting insight on how these popular branded foods fit in the spectrum of the value of the protein. This project aims to build a concise, structured data pipeline with an in-depth analytical framework that enables understanding of these products using consistent nutrition metrics. Below I will explain how this was done in this project and the entire process that brought out the answers of the goal's significance.

Data Cleaning:

This process took long to determine which data was needed and which proved ineffective for later implementations. The first step I took was to sample the data so I could effectively answer my analytical queries. I also focused on which data would better align with my analysis and provide me with a data-driven story of protein efficiency across branded foods. With this in mind, I narrowed the raw data, which was extremely large (I will cover the size in the summary), from about 3 gigabits to roughly 33 megabits, into two distinct tables that I will use going forward for the rest of the projects. The two tables I chose, which would prove to be the most informative for the project, were the food and its nutrients.

Furthermore, after I got a cleaned version of the raw data, I categorized it because I wanted to compare the protein and calorie content. I also removed any N/A values to avoid interference with the analysis and merged the two extracted tables. I also stored the protein (g) value in variable x and the calories (kcal) in variable y, without including any impossible metrics that would tamper with the query.

```

foods = pd.read_csv(
    "FoodData_Central_branded_food_csv_2026-04-30/food.csv",
    low_memory= False
)

nutrients = pd.read_csv(
    "FoodData_Central_branded_food_csv_2026-04-30/food_nutrient.csv",
    low_memory= False
)

foods.columns = foods.columns.str.strip()

protein = nutrients[nutrients['nutrient_id'] == 1003][['fdc_id', 'amount']]
calories = nutrients[nutrients['nutrient_id'] == 1008][['fdc_id', 'amount']]

nutrition = pd.merge(protein, calories, on="fdc_id")

df = pd.merge(
    foods[['fdc_id', 'description']],
    nutrition,
    on="fdc_id"
)

df = df.dropna()

df['amount_x'] = pd.to_numeric(df['amount_x'], errors='coerce')
df['amount_y'] = pd.to_numeric(df['amount_y'], errors='coerce')

df = df.dropna()

df = df[df['amount_x'] > 0]
df = df[df['amount_y'] > 0]
df = df[df['amount_x'] < 100]
df = df[df['amount_y'] < 2000]

```

Query:

For the queries I kept it pretty simple and mapped out what questions I wanted answered with the main focus of calculating trends in protein efficiency. The questions I answered include the following:

- Top 10 most protein-efficient food
- Top 10 least protein-efficient food
- How each protein and calories correlate with each other
- Average protein efficiency based on categories

Plotting:

I used scatterplots and mainly barplots to show the trends and tradeoffs of previous queries that would provide a deep understanding and visualization on the proportion of

the analysis. You can also gather more data from these plots and expand the queries even further by intently looking at the certain relationships between protein across branded foods. I wanted to initialize this in the plotting so it enables the viewer to grasp an understanding on the topic as well as gain key insights on rational data. Providing this analysis will spike curiosity in gathering more inquiries for future innovation of gathering this data. Mainly plotting these queries will communicate the data in a more visually appealing way to where someone who is not a data scientist or someone who cares about the logistics can view it as well.

Model:

I provided a linear regression model showing the trendline alongside the actual data points for the relationship between calories and protein. I decided to look at the Branded Food dataset to see if there's actually a reliable relationship between a food's calorie count and the amount of protein it provides. We see "High Protein" labels on everything now, from water to chips, so I wanted to use data to cut through the marketing and find out which categories are actually worth it, and if we can predict protein just by looking at the calorie count.

I had implemented a model error calculator that presented if the model I had created would be errors from the actual dataset. On top of those boundaries I also structured a model accuracy and R^2 system that calibrated the accuracy of the model itself.

Summary/Goal Achieved?:

I want my project to be seen as a key to understanding branded foods and to measure its success based on how well the data is cleaned and whether clear comparisons are shown. I have also presented how much cleaning I have done using the data by showing the size difference between the raw and cleaned data. The project has achieved:

- Produce a clean and useable dataset
- Create a meaningful protein-efficiency measurement
- Compare foods and categories in a way that is easy to interpret
- Use visualizations that clearly support my findings
- Explain important trends and tradeoffs instead of oversimplifying the results